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1. Which email protocols should GlobalTech Ltd. adopt for sending and receiving emails securely?

GlobalTech should use SMTP with TLS for sending emails and IMAP with SSL/TLS (IMAPS) for receiving and synchronizing emails across devices. IMAP is the better choice because it leaves the emails on the server while giving an employee access to the same mailbox across multiple, disparate devices: laptops, phones, and workstations from remote locations. POP3 is dated for modern multi-device communication because, by default, it downloads the message locally onto a device and then deletes the message from the server, making it challenging to keep messages across all devices synchronized. Using TLS/SSL will ensure that all email communications are encrypted in transit, maintaining sensitive data related to clients and reducing the possibility of intercepting or unauthorized access.

2. How can the company prevent emails from being flagged as spam?

They should apply SPF to validate authorized mail servers, DKIM for digitally signing the outgoing emails and confirming their authenticity, and DMARC for defining policies on how receiving mail servers should treat suspicious or failed authentication messages. Furthermore, they have to make sure that Reverse DNS records are set correctly, maintain a secure configuration of the mail server without open relay so as not to send unauthorized emails, and use reputation management practices, such as avoiding spam-like content, using proper email formatting, and more. Such combined measures increase the trust of the domain name and ensure that no legitimate corporate emails will be blocked or marked as spam by the external mail servers.

3. What measures can be implemented to ensure email encryption and protect sensitive data?

GlobalTech should employ TLS to encrypt email in transit and E2EE with S/MIME or PGP to ensure the actual content of highly confidential information is kept secure. Other measures include DLP tools, which help prevent accidental sharing of sensitive data, and access controls and MFA to protect email accounts from unauthorized access. Automatic email classification can be implemented, warning the sender when email recipients are external, protecting sensitive communications. TLS will provide security for the connection, while S/MIME or PGP will encrypt the actual message content.

4. How can employees safely access their emails from different devices?

Employees can securely view their emails from various devices through IMAP over SSL/TLS for secure authentication and synchronization. MDM enforces security policies on phones and tablets, while VPN access protects connections when working remotely. Additional measures to protect accounts and data include Multi-Factor Authentication, strong password policies with rotation, encryption of devices, and remote wipe. Company-approved email applications, like Microsoft Outlook or other secure mail apps, ensure that emails remain synchronized, encrypted, and protected on all devices.

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